Hellenic Complex Systems Laboratory

Inferences about the Difference between Two Proportions

Technical Report XVIII

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2018

Inferences about the Difference between Two Proportions

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Short Description of the Demonstration

This Demonstration explores statistical inference for the difference between two population proportions associated with the presence of a condition or trait. It computes the statistical significance and confidence interval for the difference between the two proportions and plots the corresponding confidence limits as functions of the selected confidence level. The calculations are performed for different counts of individuals with and without the condition, and for different confidence levels used in interval estimation.

Search Terms

proportions, difference between proportions, confidence interval, inference, critical ratio statistic, statistical significance, statistics

first population

population with the condition

90

population without the condition

10

second population

population with the condition

850

population without the condition

150

confidence level 1–α

0.95

first proportion	0.900
second proportion	0.850
point estimate of the difference	-0.050
significance level α of the difference	0.114399

95.% confidence interval for the difference	
lower confidence limit	-0.110
upper confidence limit	0.010

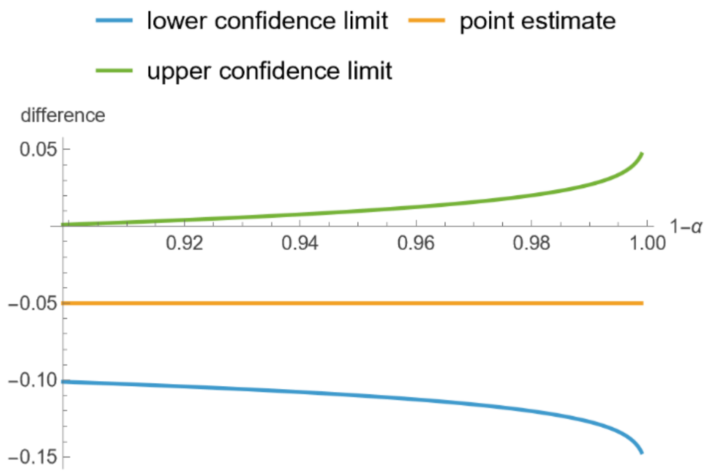


Figure 1*: Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus* confidence level for the difference*, with the settings shown at the left.*

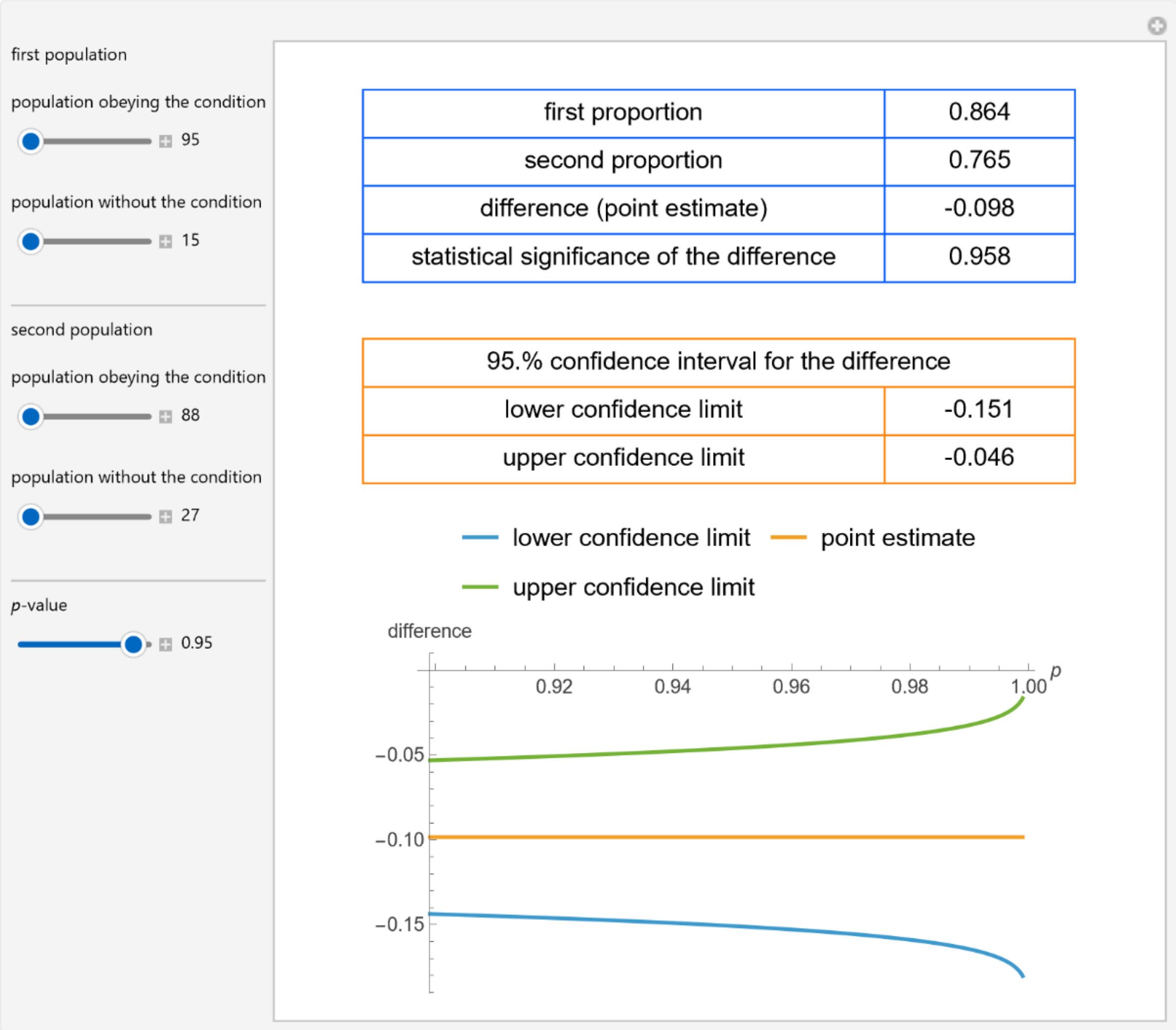


Figure 2: *Point estimate and 99% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus* confidence level for the difference*, with the settings shown at the left.*

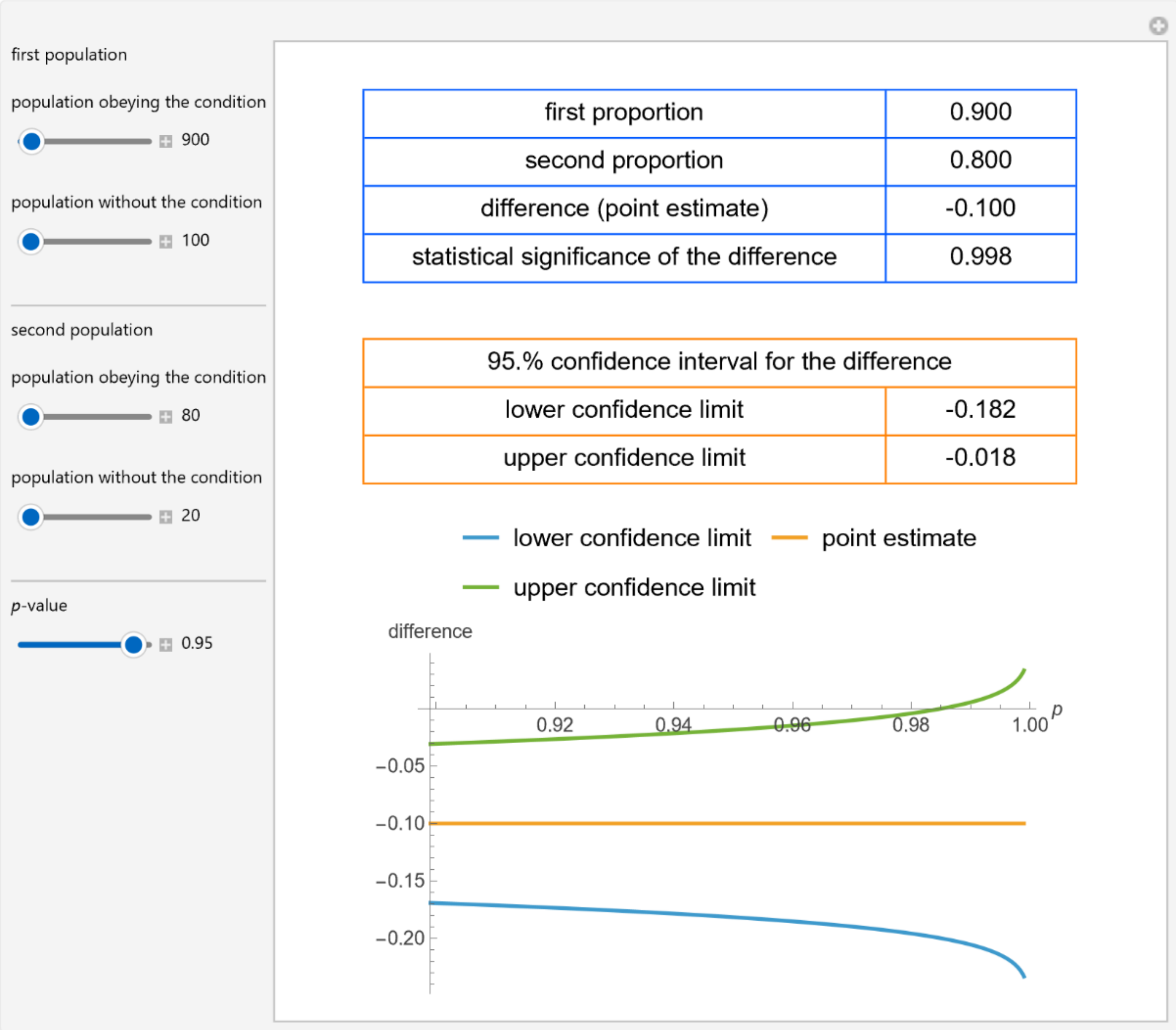
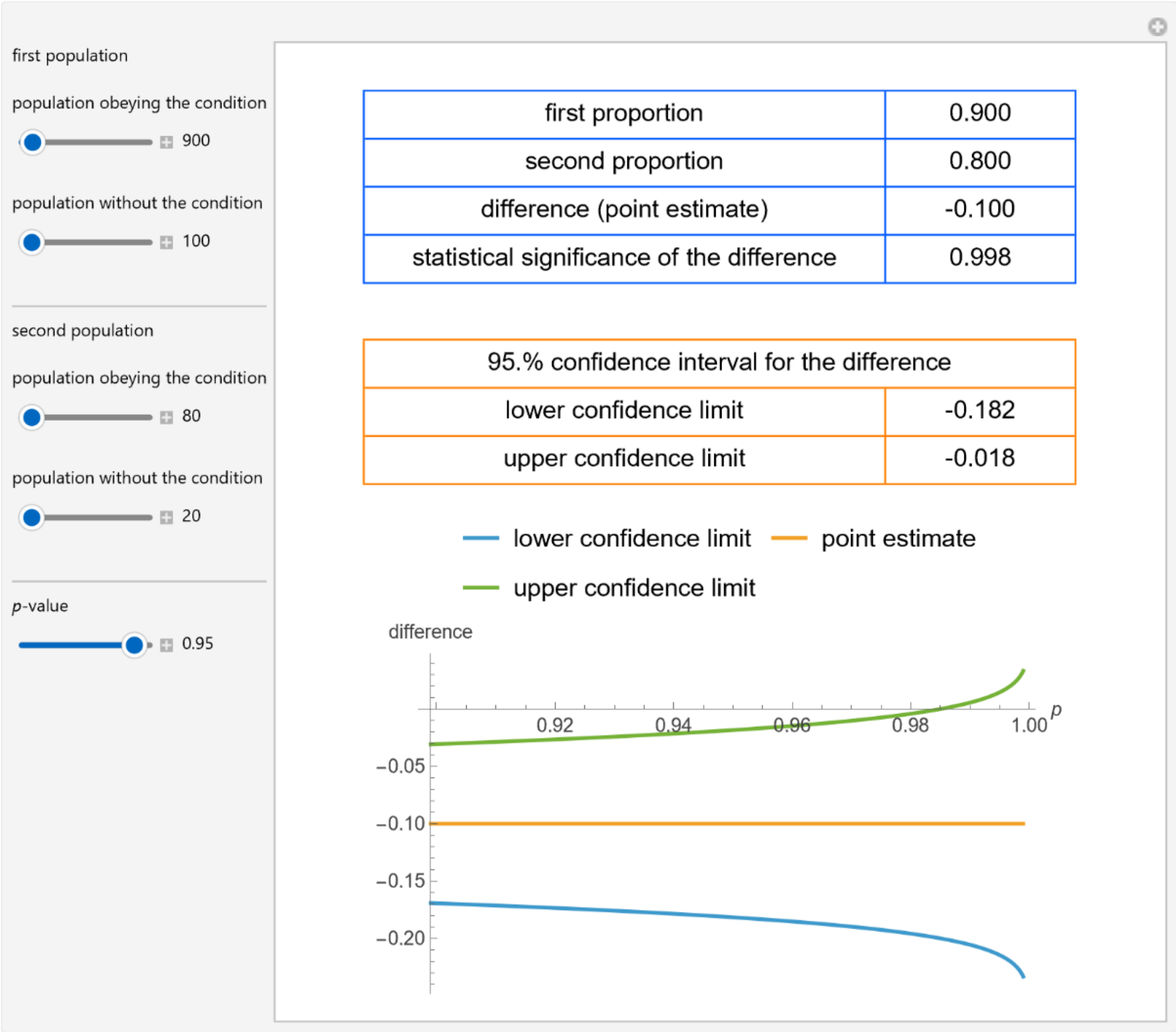


Figure 3*: Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus* confidence level for the difference*, with the settings shown at the left.*



first proportion	0.900
second proportion	0.800
difference (point estimate)	-0.100
statistical significance of the difference	0.998

95.% confidence interval for the difference	
lower confidence limit	-0.182
upper confidence limit	-0.018

Figure 4*: Point estimate and 95% confidence interval for the difference of two proportions of populations with a condition, as well as their plots versus* confidence level for the difference*, with the settings shown at the left.*

Details

The significance level α of the difference between the two proportions is assessed by means of the z-score (or critical ratio statistic) [1]. The confidence interval for the difference between the proportions at a confidence level 1-α is calculated as in [2].

References

[1] J. L. Fleiss, B. Levin and M. C. Paik. Methods for Generating a Fourfold Table. *Statistical Methods for Rates and Proportions*, 3rd ed., Hoboken, NJ: J. Wiley, 2003 pp. 51–52.

[2] J. L. Fleiss, B. Levin and M. C. Paik. A Simple Confidence Interval for the Difference between Two Independent Proportions. *Statistical Methods for Rates and Proportions*, 3rd ed., Hoboken, NJ: J. Wiley, 2003 pp. 60–61.

Demonstration

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Source

Programming language: Wolfram Language
Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/InferencesAboutTheDifferenceBetweenTwoProportions.nb

First published in 2018 as part of the Wolfram Demonstrations Project:
https://demonstrations.wolfram.com/InferencesAboutTheDifferenceBetweenTwoProportions/

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS
Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.
System memory (RAM): 4 GB+ recommended.

Citation:

Chatzimichail T, Hatjimihail AT. *Inferences about the Difference between Two Proportions.* Ver 1.1.2. Hellenic Complex Systems Laboratory; 2026.

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Citation

Chatzimichail T, Hatjimihail AT. *Inferences about the Difference between Two Proportions.* Technical Report XVIII. Hellenic Complex Systems Laboratory; 2018.

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First Published: August 2, 2018

Revised: May 23, 2026